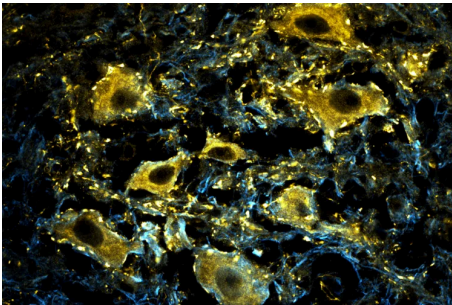


[Home \(/\)](#) > [About \(/about\)](#) > [News \(/about/news\)](#)

> New publication: Synaptopathy in the TDP-43 Δ NLS Mouse Model of Sporadic Amyotrophic Lateral Sclerosis

New publication: Synaptopathy in the TDP-43 Δ NLS Mouse Model of Sporadic Amyotrophic Lateral Sclerosis



Nov 2025: St Andrews ALS research team finds early synapse loss in ALS mouse model

Average reading time 2 mins

A new study from University of St Andrews researchers shows that specific brain connections weaken early in a mouse model of ALS, offering fresh clues to how the disease begins. The team is supervised by Dr Matthew Broadhead and Prof Gareth Miles at University of St Andrews.

The following summary is provided by senior author Dr Matthew Broadhead, University of St Andrews.

The deterioration of movement seen in patients with Amyotrophic Lateral Sclerosis (ALS) is caused by the degeneration of motoneurons in the brain and spinal cord.

One of the reasons motoneurons degenerate is that the inputs they receive from other neurons called "synapses" become dysfunctional or degenerate. However, neurons receive such a wide and diverse array of different synaptic inputs from neurons throughout the nervous system - how do we know which ones are vulnerable in the disease?

This research used high-resolution fluorescence microscopy to reveal specific types of synapses that are vulnerable in a model of ALS. This will provide ALS researchers with a better understanding of which types of synaptic connections are vulnerable and may help identify potential therapeutic targets in the disease.

We use cookies on this site to enhance your user experience

Relevant links

Accept

No, thanks

By clicking the Accept button, you agree to us doing so. [More info](#)

Read the scientific paper DOI: 10.1111/ejn.70320: [Wiley Online Library](https://onlinelibrary.wiley.com/doi/10.1111/ejn.70320).
(<https://onlinelibrary.wiley.com/doi/10.1111/ejn.70320>).
(external website)

[Prof Gareth Miles \(/people/prof-gareth-b-miles\)](/people/prof-gareth-b-miles) profile

[Dr Matthew Broadhead \(https://research-portal.st-andrews.ac.uk/en/persons/matthew-broadhead/\)](https://research-portal.st-andrews.ac.uk/en/persons/matthew-broadhead/) profile
(external website)

Funding

This research was funded by Tenovus Scotland, RS Macdonald Charitable Trust and The Lundbeck Foundation

Image credit: Dr Matthew Broadhead

This article was published on: Monday, 24 November, 2025

We use cookies on this site to enhance your user experience

By clicking the Accept button, you agree to us doing so.